

12(a)	probability of decay (of a nucleus)	M1
	per unit time	A1
12(b)	$A = \lambda N$	C1
	$N = \text{mass} / (\text{nucleon number} \times u)$	C1
	$2.92 \times 10^9 = (\lambda \times 5.87 \times 10^{-10}) / (131 \times 1.66 \times 10^{-27})$	A1
	$\lambda = 1.08 \times 10^{-6} \text{ s}^{-1}$	
12(c)	- sample emits radiation in all directions - some radiation is absorbed by air/detector window - self-absorption within the source - dead time/inefficiency of detector <i>Any two points, 1 mark each</i>	B2